

HARDWARE AND COMPONENTS BASICS

DAY-5 Assemble and Disassemble a PC Practical

1. Internal Desktop PC Components & Functional Summary

- **Motherboard (Main Circuit Board):** Serves as the central communication backbone linking the processor, memory, storage devices, and bus expansion interfaces.
- **CPU (Microprocessor):** Executes software instructions, carries out arithmetic calculations, and manages core hardware system logic.
- **CPU Cooler & Fan Assembly:** Regulates processor operating temperature by dissipating heat buildup across aluminum cooling fins with an active airflow fan.
- **RAM (System Memory):** Provides high-speed volatile workspace for caching running applications, background services, and current system tasks.
- **PSU (Power Supply Unit):** Converts external 230V alternating current (AC) into regulated low-voltage direct current (DC) rails (+12V, +5V, +3.3V).
- **Hard Disk Drive (HDD / SSD Storage):** Provides non-volatile magnetic/flash storage for the installed OS environment, applications, drivers, and user files.
- **Dedicated GPU (Graphics Card):** Specialized expansion unit with integrated video memory engineered to compute and render real-time display graphics.
- **Cabinet Ventilation Fans:** Maintains directed airflow circulation inside the chassis to expel hot air and pull in ambient cool air.
- **Front I/O & Control Headers:** Wiring harness routing chassis power/reset switches, drive status LEDs, and front USB/audio jacks directly to board pins.
- **PCIe Expansion Slots:** Peripheral sockets allowing installation of auxiliary sound cards, Wi-Fi modules, and additional connectivity hardware.

2. Complete PC Disassembly Procedure (Step-by-Step)

Step 1: AC Mains & Peripherals Disconnection

Turned off wall power and detached the primary 3-pin AC power cord, display monitor lead, and USB peripherals from rear ports.



Step 2: Chassis Side Panel Removal

Loosened rear chassis thumbscrews with a cross screwdriver and smoothly unhooked the side cover panel from its frame slots.



Step 3: Internal Power Harness & Data Cable Detachment

Released plastic latching clips on the 24-pin ATX motherboard feed, 4-pin CPU line, and SATA drives before decoupling wires.



Step 4: Memory Module (RAM) Extraction

Pressed downward on the dual ejector retention clips bordering the slot to let the RAM stick pop free without touching gold pins.



Step 5: Hard Disk Drive Removal

Removed retaining fasteners securing the storage tray and slid the mechanical 3.5-inch drive out of the internal drive bay cage.



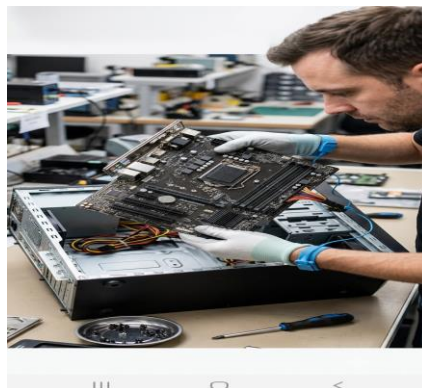
Step 6: Dedicated Video Card (GPU) Detachment

Unscrewed the metal PCI bracket screw, unlocked the PCIe slot retention tab, and eased the graphics card straight out



Step 7: Motherboard Removal from Standoffs

Systematically backed out all brass standoff chassis screws and carefully lifted the motherboard free of the interior chamber.



Step 8: Organized Sorting & Bench Storage

Arranged all decoupled hardware items safely across a clean surface with screws segregated in a compartmentalized container.



3. Step-by-Step PC Reassembly & Sequence Justification

Step 1: Motherboard Seating & Alignment

Positioned the motherboard into the chassis against the rear I/O shield and secured it over all grounding brass standoffs.



Step 2: RAM Module Installation

Aligned key notches on the memory module with the slot division and pushed down until dual locking clips snapped shut.



Step 3: Storage Unit Installation

Slid the storage hard disk into the mounting cage and fastened lateral screws to ensure mechanical rigidity.



Step 4: Dedicated Graphics Card Installation

Aligned the PCIe card with the primary slot until the latch locked, then anchored the external PCI metal bracket.



Step 5: Cable Routing & Power Connections

Hooked up the main 24-pin ATX plug, auxiliary CPU line, SATA data/power lines, and cabinet front panel switch leads.



Step 6: CPU Cooling Unit Verification

Verified solid heatsink contact over processor thermal compound and coupled the 4-pin PWM lead onto the 'CPU_FAN' header.



Step 7: Enclosure Closing

Tucked stray cables into side channels, aligned the exterior side panel along tracking rails, and tightened the rear thumbscrews.



Step 8: Peripheral Hookup & Hardware Post-Check

Connected the AC power line and monitor, powered the computer on, and loaded the BIOS utility to verify all hardware detection.



4. Pre-Work Safety Guidelines & Hardware Tool Checklist

Checklist of essential tools and safety protocols required prior to starting hardware operations:

- **Mains Power Isolation:** Always switch off and physically unplug the AC power cord from the wall to eliminate electric shock hazards entirely.
- **Capacitor Residual Bleeding:** Hold down the PC power button for several seconds following power cable removal to drain stored charge in PSU capacitors.
- **ESD Wrist Band Usage:** Wear a grounded anti-static wrist strap to prevent electrostatic discharge from damaging microscopic silicon IC circuits.
- **Phillips Head #2 Screwdriver:** Use the standard crosshead tool to drive casing, motherboard, and bay screws without stripping screw slots.
- **Magnetic Sorting Dish:** Keep chassis, PSU, and internal drive screws separated by type to prevent lost hardware falling into open components.
- **Edge-Only Component Handling:** Hold PCBs, RAM sticks, and expansion cards strictly by non-conductive edges to avoid skin moisture contaminating contacts.
- **Zero Excessive Mechanical Force:** Keyed connectors (RAM notch, PCIe slots, SATA plugs) fit in one orientation; forcing them breaks socket pins.
- **Clean, Non-Conductive Workbench:** Operate on a solid, dry wooden or anti-static mat free from conductive clutter, loose screws, or static-generating fabrics.
- **Strict Liquid Isolation:** Keep beverages and water bottles away from the workspace to eliminate risks of catastrophic short circuits.
- **Pre-Boot Connection Audit:** Carefully inspect every power lead, data connector, and fan header before engaging the main power switch.